Mathematical tools

Learning goals and Literature

After this set of video lectures you should...

- Know some important mathematical definitions and results that you need to be able to follow the course TTK4150 Nonlinear Control Systems.

The video lectures are based on

Khalil	Appendix A (Chapter 5.1)

A mathematical toolbox

A crash course in Linear methods...

- Metrics
- Norms
  - The set of n-dimensional vectors and p-norms
  - The set of real-valued continuous functions, corresponding norms and the $\mathscr{L}_p$-space
  - Matrix norms
- Inner Products
  - Relations between inner products and norms
    - Schwarz' inequality
    - Hölder's inequality
- Differentiable functions

Metric, Norm, Inner product

Space

- A set X + a topological structure

Set

Example

- $X = \mathbb{R}^n$
- $X = \mathbb{C}[a,b]$

Topological structure

	metric	norm	inner product
	$\mathrm{d}(,)$	$\|\cdot\|$	$<,>$
Set	Metric space	Banach space	Hilbert space

Metric (Distance function)

Definition

A real-valued function $d(x,y)$ is a metric on X iff (if and only if) $\forall x,y,z \in X$

M1) Positive

$$d(x,y) \geq 0$$
$$d(x,x) = 0$$

M2) Strictly positive

$$d(x,y) = 0 \Rightarrow x = y$$

M3) Symmetry

$$d(x,y) = d(y,x)$$

M4) Triangle inequality

$$d(x,y) \leq d(x,z) + d(z,y)$$

`http://www.khanacademy.org:`
`"triangle inequality theorem"`

Metric

Metrics on $\mathbb{R}^n$:

where $x = [x_1, \ldots, x_n]^T$, $y = [y_1, \ldots, y_n]^T$ and $x_i, y_i \in \mathbb{R}\ \forall i = 1, \ldots, n$

$$d_1(x,y) = |x_1 - y_1| + \cdots + |x_n - y_n|$$

$$d_2(x,y) = \sqrt{|x_1 - y_1|^2 + \cdots + |x_n - y_n|^2}$$

$$d_p(x,y) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{\frac{1}{p}}, \quad p \in [1, \infty]$$

$$d_\infty(x,y) = \lim_{p \to \infty} d_p(x,y) = \max_i |x_i - y_i|$$

Metric

Metrics on $C[a,b]$:

Where $f,g:[a,b]\to\mathbb{R}$ are continuous functions

$$d_p(f,g)=\left(\int_a^b |f(\tau)-g(\tau)|^p\,d\tau\right)^{\frac{1}{p}},\quad p\in[1,\infty]$$

$$d_\infty(f,g)=\lim_{p\to\infty} d_p(f,g)=\max_{a\le t\le b}|f(t)-g(t)|$$

Metric spaces

$(\mathbb{R}^n, d_p(x,y))$ and $(C[a,b], d_p(f,g))$
are Metric spaces

Norm (Concept of length/size)

Definition

A real-valued function $\|x\|$ is a norm on X iff $\forall x, y \in X$

N1) Positivity

$$\|x\| \geq 0$$

N2) Triangle inequality

$$\|x+y\| \leq \|x\| + \|y\|$$

N3) Homogeneity

$\|\alpha x\| = |\alpha| \|x\|$ for arbitrary scalar α

N4) Positive definiteness

$$\|x\| = 0 \Leftrightarrow x = 0$$

Induced metric

Norm $\overset{Induces}{\rightarrow}$ Metric

- $d(x,y) = \|x-y\|$ is a metric

Norms on $\mathbb{R}^n$

Norms on $\mathbb{R}^n$

$$\left.\begin{aligned} \|x\|_p &= \left(\sum_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}}, \quad p \in [1,\infty] \\ \|x\|_\infty &= \max_i |x_i| \end{aligned}\right\} p-\text{norms}$$

Banach space

$(\mathbb{R}^n, \|\cdot\|_p)$ is a Banach space

- $\mathbb{R}^n$: All p-norms are equivalent:

  $\forall \alpha, \beta \in \mathbb{N}_+ \; \exists \; c_1, c_2 \in \mathbb{R}_+ : \; c_1 \|x\|_\alpha \leq \|x\|_\beta \leq c_2 \|x\|_\alpha$

Norms on $C[a,b]$

$$\left.\begin{aligned} \|f\|_{\mathscr{L}_p} &= \left(\int_a^b |f(\tau)|^p d\tau\right)^{\frac{1}{p}}, \quad p \in [1,\infty] \\ \|f\|_{\mathscr{L}_\infty} &= \sup_{a \le t \le b} |f(t)| \end{aligned}\right\} \quad \mathscr{L}_p - \text{norms}$$

$\mathscr{L}_p$-space

$$(C[0,\infty), \mathscr{L}_p - \text{norm}) \text{ is a Banach space}$$

- $f \in \mathscr{L}_p \Leftrightarrow \|f\|_{\mathscr{L}_p}$ is bounded, i.e. $\exists\, c : \|f\|_{\mathscr{L}_p} \le c$

Inner product (Concept of angle)

Definition

A real-valued function $<x,y>$ is an inner product on X iff $\forall\, x,y,z \in X$

IP1) Additivity

$$<x+y,z>=<x,z>+<y,z>$$

IP2) Homogeneity

$<\alpha x,y>=\alpha <x,y>$ for arbitrary scalar α

IP3) Symmetry

$$<x,y>=\overline{<y,x>}$$

IP4) Positivity

$<x,x>>0$ if $x\neq 0$

Schwarz' inequality

Induced norm

Inner product $\stackrel{Induces}{\rightarrow}$ Norm

- $\|x\| = < x, x >^{\frac{1}{2}}$

A useful property (holds only for the induced norm):

Schwarz' inequality

Let $< x, y >$ be an inner product on a set X. Then

$$|< x, y >| \leq \|x\| \cdot \|y\|$$

when $\|\cdot\|$ is defined as above.

Inner products

Inner product on $\mathbb{R}^n$

$< x, y > \triangleq x^T y = \sum_{i=1}^n x_i y_i$

$$\overset{Induces}{\longrightarrow} \|x\| =< x, x >^{\frac{1}{2}} = \left(\sum_{i=1}^{n} x_i^2 \right)^{\frac{1}{2}}$$ Euclidean norm

Inner product on $C[a,b]$

$< f, g > \triangleq \int_a^b f(\tau) g(\tau) d\tau$

$$\overset{Induces}{\longrightarrow} \|f\| =< f, f >^{\frac{1}{2}} = \left(\int_a^b |f(\tau)|^2 d\tau \right)^{\frac{1}{2}}$$ $\mathscr{L}_2$ norm

Hölder's inequality

$\mathbb{R}^n$ case

$$\forall x, y \in \mathbb{R}^n : \quad \sum_{i=1}^{n} |x_i y_i| \leq \|x\|_p \cdot \|y\|_q, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

$\mathscr{L}_p$ case

$$\forall f \in \mathscr{L}_p, g \in \mathscr{L}_q : \int_a^b |f(\tau) g(\tau)| d\tau \leq \|f\|_p \cdot \|g\|_q, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

Matrix norms (i.e. norms on $\mathbb{R}^{m\times n}$)

Definition

If $A \in \mathbb{R}^{m\times n}$ we have

$$\|A\|_p \triangleq \sup_{x\neq 0} \frac{\|Ax\|_p}{\|x\|_p} = \max_{\|x\|_p=1} \|Ax\|_p \quad p \in [1,\infty]$$

$$\downarrow$$

$$\|A\|_1 = \max_j \sum_{i=1}^{m} |a_{ij}|$$

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)}$$

$$\|A\|_\infty = \max_i \sum_{j=1}^{n} |a_{ij}|$$

Matrix norms

Useful properties

$$\|Ax\|_p \leq \|A\|_p \cdot \|x\|_p$$

$$\frac{1}{\sqrt{n}}\|A\|_\infty \leq \|A\|_2 \leq \sqrt{m}\|A\|_\infty$$

$$\frac{1}{\sqrt{m}}\|A\|_1 \leq \|A\|_2 \leq \sqrt{n}\|A\|_1$$

$$\|A\|_2 \leq \sqrt{\|A\|_1\|A\|_\infty}$$

$$\|AB\|_p \leq \|A\|_p\|B\|_p, \quad A \in \mathbb{R}^{m\times n},\ B \in \mathbb{R}^{n\times l}$$

Differentiable functions

Notation

$f : \mathbb{R}^n \to \mathbb{R}^m$

C^0 at x_0: f_i continuous at x_0 $\quad \forall i = 1, \ldots, m$

in S: f_i continuous in S $\quad \forall i = 1, \ldots, m$

C^1 $\quad f_i(x)$ continuous $\quad \forall i = 1, \ldots, m$

$\frac{\partial f_i}{\partial x_j}$: exists and continuous $\quad \forall i = 1, \ldots, m, \; j = 1, \ldots, n$

C^2

$\downarrow$

C^∞ $\quad$ "smooth"

Differentiable functions cont.

Jacobian

$f : \mathbb{R}^n \to \mathbb{R}^m$

$$\frac{\partial f}{\partial x} \triangleq \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

Gradient

Special case for $m = 1$ (scalar functions) $h : \mathbb{R}^n \to \mathbb{R}$

$$\frac{\partial h}{\partial x} = \left[\frac{\partial h}{\partial x_1} \cdots \frac{\partial h}{\partial x_n} \right]$$

$$\nabla h \triangleq \begin{bmatrix} \frac{\partial h}{\partial x_1} \\ \vdots \\ \frac{\partial h}{\partial x_n} \end{bmatrix} = \left(\frac{\partial h}{\partial x} \right)^T$$